

Requirement Analysis

Solution Requirements

Team ID	SWTID-2026-2971
Team Leader	Gowsik Raja S
Team Members	Shrija M; Swetha M; Sivaram A

1. Functional Requirements

ID	Requirement	Priority
FR-01	Prepare a spreadsheet containing employee data for import.	Required
FR-02	Create Employee Test (u_employee_test) with Employee ID, Employee Name, Email, Department, and Location as String fields.	Required
FR-03	Create Employee Import (u_employee_import) as the Import Set staging table.	Required
FR-04	Create Sample Spreadsheet Import Transform Map from Employee Import to Employee Test.	Required
FR-05	Use Auto Map Matching Fields and Mapping Assist to establish mappings.	Required
FR-06	Execute the transform and verify successful record creation.	Required
FR-07	Configure Coalesce on a field map to prevent duplicate records and support updates.	Required
FR-08	Create Employees by Department, Employees by Location, and Employee List Report.	Required
FR-09	Create Employee Analytics Dashboards and add all three reports.	Required

2. Non-Functional Requirements

- The workflow should be understandable and repeatable for structured spreadsheet imports.
- Mappings should be explicit and reviewable before transformation.
- Import outcomes should be verifiable through transform history and target-table records.
- Reporting should provide simple visibility into the imported data.

3. Acceptance Summary

The requirements are considered satisfied when the documented import, transformation, Coalesce, reporting, and dashboard steps can be executed as described in the project guide.

4. Detailed Functional Requirements

The requirements cover the full lifecycle rather than only the upload action. They include preparation of source data, creation of the target table, creation of the staging table, transformation mapping, execution, validation, matching behavior, reporting, and dashboard presentation. Each stage is dependent on the successful completion of the preceding stage.

5. Data Requirements

The target Employee Test table contains five documented String fields: Employee ID, Employee Name, Email, Department, and Location. The import process receives spreadsheet data and stages it in Employee Import before the Transform Map moves the information into the target table.

6. Business Rules Represented by the Workflow

A newly encountered employee identifier can result in an inserted target record. When Coalesce matching identifies an existing record, changed values can update that record instead of producing a second record. When the same data is imported again, the documented test shows the rows being ignored.

7. Requirement Traceability

FR-01 and FR-02 establish source and target data structures. FR-03 through FR-06 implement the import pipeline. FR-07 addresses repeat-import behavior. FR-08 and FR-09 provide reporting and dashboard visibility. This traceability keeps the requirements aligned with the actual steps demonstrated in the source guide.